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International Mathematics Assessments for Schools

2025 / 2026 JUNIOR DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2025 / 2026 JUNIOR DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. On an analogue clock, the hour hand is 5 cm long and the minute hand is 10 cm long. It is currently 6:00. After x minutes, the hour and minute hands will form two of the three sides of a right triangle (where the third side is the line segment that connects the tips of the hour and minute hands). If $0 < x < 60$, how many possible values of x are there?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

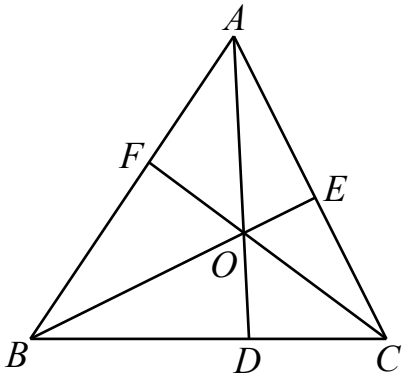
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2. Alice, Betty, Carl, David and Elva attend the same school. When walking alone, each person moves at a speed of 2.0 m per second. When walking in a group, their speed depends on the group size: any pair walks at 1.5 m/s, any trio at 1.0 m/s and any quartet at 0.5 m/s. One day, Alice and Betty leave the school together at 16:00:00, Carl at 16:01:00, David at 16:02:00 and Elva at 16:03:00. They all walk in the same direction along a straight road and whenever someone catches up with another, they join the group immediately. How far, in m, from the school will they be when all five of them are together?

(A) 520 (B) 480 (C) 360 (D) 240 (E) 120

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3. As shown in the diagram below, ABC is a triangle, where AD, BE and CF are drawn from the vertices such that all three lines intersect at point O. If $\frac{EO}{EB} = \frac{1}{4}$ and $\frac{FO}{FC} = \frac{2}{5}$, what is the value of $\frac{DO}{DA}$?



(A) $\frac{13}{40}$ (B) $\frac{7}{20}$ (C) $\frac{3}{8}$ (D) $\frac{3}{5}$ (E) $\frac{17}{40}$

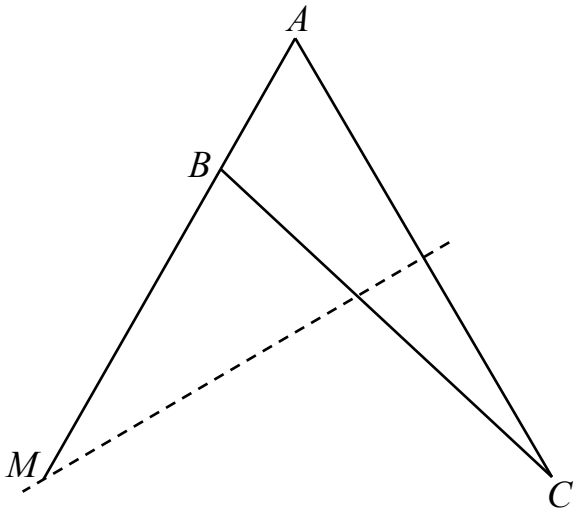
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4. What is the maximum number of integers that can be selected from the twenty numbers {1, 2, 3, …, 20} such that the sum of any two distinct selected integers is not included among the selected integers?

(A) 9 (B) 10 (C) 11 (D) 12 (E) 13

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5. As shown in the diagram below, ABC is a triangle where $\angle ABC > \angle BAC = 60°$, $AB = 1$ cm and $AC = 2\sqrt{3}$ cm. If the perpendicular bisector of AC intersects line AB at point M, what is the length, in cm, of MB?

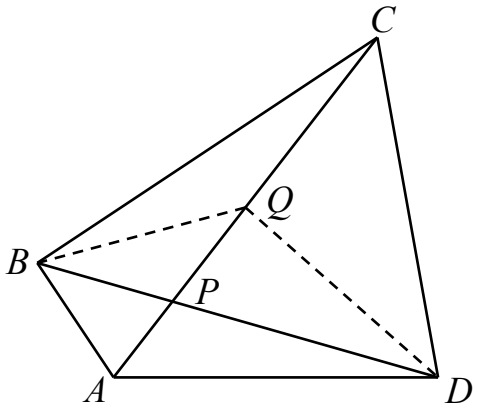


(A) $\sqrt{3}$ (B) 3 (C) $2\sqrt{3}-1$ (D) $2\sqrt{3}-2$ (E) $2\sqrt{6}$

6. Which number from the following options is always a prime factor of $p^{p-1}+2$, for any prime number $p \geq 5$?

(A) 3 (B) 7 (C) 11 (D) 13 (E) 19

7. As shown in the diagram below, $ABCD$ is a convex quadrilateral with diagonals intersecting at point P. The sum of the areas of triangles PAB and PCD equals the sum of the areas of triangles PAD and PCB. If Q is the midpoint of AC and Q does not coincide with P, what is the value of $\dfrac{BP}{DP}$?

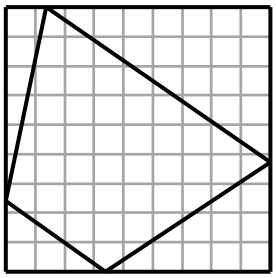


(A) $\dfrac{1}{2}$ (B) $\dfrac{2}{5}$ (C) $\dfrac{3}{5}$ (D) $\dfrac{2}{3}$ (E) 1

8. How many integers N between 50 and 120, inclusive, can be expressed as the sum of two or more consecutive positive integers?

(A) 67 (B) 68 (C) 69 (D) 70 (E) 71

9. Each vertex of a convex quadrilateral lies on a different side of a $9\text{ cm} \times 9\text{ cm}$ square, as shown in the diagram. What is the minimum possible perimeter, in cm, of the quadrilateral?



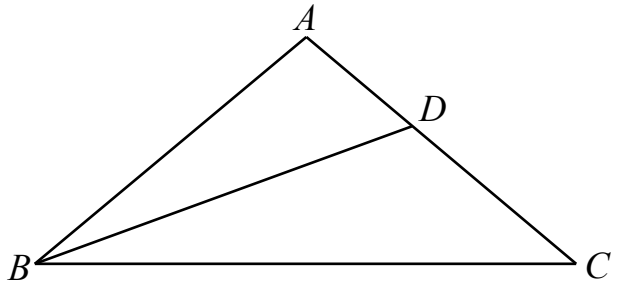
(A) $12\sqrt{5}$ (B) $18\sqrt{2}$ (C) $18\sqrt{3}$ (D) $4\sqrt{41}$ (E) $27\sqrt{2}$

10. There are 100 distinct points on a line. The midpoint of every segment determined by any two of these points is painted red. What is the minimum number of red points?

(A) 1 (B) 99 (C) 197 (D) 198 (E) 4950

Questions 11-20, 4 marks each

11. As shown in the diagram below, ABC is a triangle where $AB = AC$ and the angle bisector of $\angle ABC$ intersects AC at D. If $BC = BD + AD$, what is the measure, in degrees, of $\angle CBD$?



(A) 15° (B) 20° (C) 24° (D) 30° (E) 40°

12. What is the minimum possible value of $\dfrac{5-4x+x^2}{2-x}$ for all real numbers $x < 2$?

(A) $\dfrac{1}{2}$ (B) 1 (C) 2 (D) $\dfrac{13}{6}$ (E) $\dfrac{5}{2}$

13. A sequence consists of six distinct positive integers in strictly increasing order such that each term (excluding the first) is a multiple of the preceding term. If the sum of all six terms is 111, what is the smallest possible value of the largest term of the sequence?

（A）88　（B）80　（C）72　（D）64　（E）56

14. Let the number N be $\dfrac{1^4+2024^4+2025^4}{1^2+2024^2+2025^2}$. What is the remainder when N is divided by 2026?

（A）0　（B）3　（C）8　（D）2024　（E）2025

15. Alex has a 10-faced die with the faces labelled from 1, 2, …, 10. The probability of each face appearing is proportional to its number, that is, the corresponding probabilities are in the ratio 1:2:3:4:5:6:7:8:9:10. If Alex rolls the die twice, what is the probability that the sum of the two rolls is 9?

（A）$\dfrac{9}{55}$　（B）$\dfrac{12}{55}$　（C）$\dfrac{24}{55}$　（D）$\dfrac{16}{605}$　（E）$\dfrac{24}{605}$

16. There are k positive integers, not necessarily distinct, and one of which is 26. If the sum of their reciprocals is 1, what is the minimum possible value of k?

（A）3　（B）4　（C）5　（D）14　（E）26

17. There are k distinct positive integers arranged in a circle such that product of any two adjacent integers is less than 2026. What is the maximum possible value of k?

（A）91　（B）90　（C）89　（D）88　（E）87

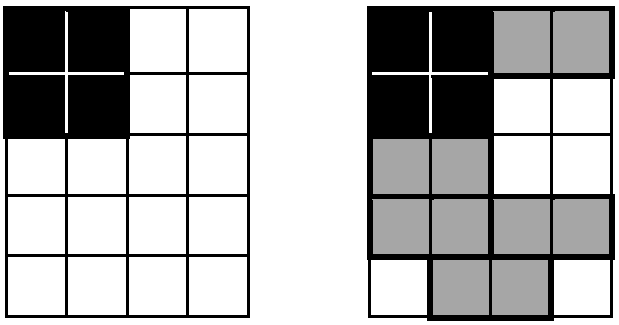
18. What is the sum of all positive integers $n \le 200$ such that the sum of all distinct positive divisors of n equals $2n-1$?

（A）126　（B）127　（C）253　（D）254　（E）255

19. A perfect square greater than 1 is called a "***good number***" if it can be expressed as a sum of consecutive positive integers starting from 1. For example, 36 is a good number since $36=6^2=1+2+3+4+5+6+7+8$. Including 36, how many *good numbers* are less than or equal to 2025?

（A）1　（B）2　（C）3　（D）4　（E）5

20. Consider a 5×4 grid, in which a 2×2 black square has already been placed in the top-left corner, as shown in the diagram below on the left. You are to place five 1×2 rectangles onto the remaining unoccupied cells. The rectangles must all be placed horizontally (no rotations allowed), aligned along the grid lines, and cannot overlap with each other or placed over any of the black squares. One valid arrangement is shown below on the right. How many such arrangements, including the given valid arrangement, are possible?



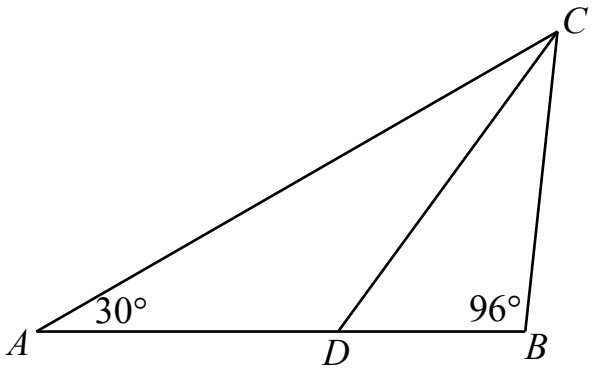
(A) 114 (B) 120 (C) 126 (D) 132 (E) 144

Questions 21-25, 6 marks each

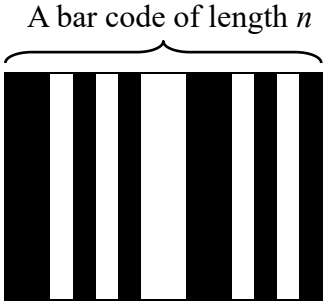
21. How many ordered pairs of real numbers (x, y), with $0<x<1$ and $0<y<1$, are there such that both $3x+5y$ and $7x+2y$ are integers?

22. Let a and b be positive integers such that $ab-a+3b=2026$, what is the minimum possible value of $a+3b$?

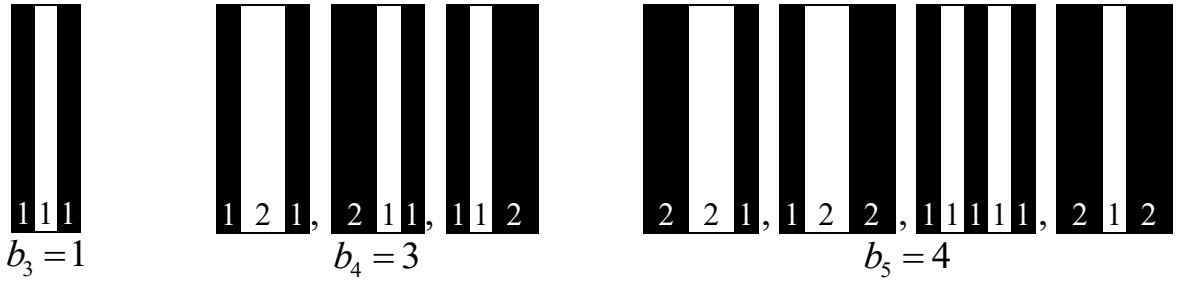
23. In triangle ABC, point D lies on AB, such that $AD=BC$. If $\angle ABC=96°$ and $\angle BAC=30°$, what is the measure, in degrees, of $\angle ACD$?



24. A barcode starts and ends with a black stripe, and consists of alternating black and white stripes. One example of a barcode is shown in the diagram below. It is read from left to right, and each stripe (whether black or white) has a width of 1 mm or 2 mm.



Let b_n be the number of barcodes of length $n \in \mathrm{N}$. If we have $b_3 = 1$, $b_4 = 3$ and $b_5 = 4$, what is the value of b_{14}?



25. A byte consists of 8 binary bits, each of which is either 0 or 1. Among all $2^8 = 256$ possible byte patterns, how many satisfy the condition that no three consecutive bits have the same value? For example, 00110101 satisfies the condition, but 0101**000**1 and 0**111**0010 do not.

* * *